Global Volatility Spillovers with the `BVAR` Package

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Main Results

Project Overview

What is our project?

We study how volatility shocks propagate across six major equity markets using a Bayesian VAR, and we compare calm and crisis regimes to see whether global transmission intensifies under stress.

Empirical design

- Six implied-volatility indices in levels
- Full sample plus calm-vs-crisis comparison
- Generalized IRFs, GFEVDs, and conditional forecasts as the main outputs

Why this is a good `BVAR` application

- Volatility is persistent and strongly comoving.
- A six-variable VAR with five lags already has 186 coefficients.
- Bayesian shrinkage is therefore central to making the model usable.

Our angle

We use this project to show how the `BVAR` package turns a theoretically rich model into an applied workflow that is easy to estimate, diagnose, and interpret.

Why BVAR?

Dimensionality problem

A VAR with M variables and p lags has

$$M + M^2 p$$

slope parameters before the covariance matrix.

With $M = 6$ and $p = 5$, this already implies

$$6 + 6^2 \cdot 5 = 186$$

coefficients to regularise.

Why Bayesian shrinkage helps

- It shrinks weak coefficients toward sensible benchmarks.
- It reduces overfitting in a medium-size system.
- It improves stability for forecasting and impulse analysis.

Practical message

For this kind of volatility system, Bayesian regularisation is not just elegant theory; it is what makes the VAR empirically tractable.

Compact Theory: VAR and Minnesota Prior

Bayesian VAR

$$\boldsymbol{y}_t = \boldsymbol{a}_0 + A_1 \boldsymbol{y}_{t-1} + \cdots + A_p \boldsymbol{y}_{t-p} + \boldsymbol{\varepsilon}_t, \qquad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\boldsymbol{0}, \Sigma)$$

Minnesota prior

$$\mathbb{E}[(A_s)_{ij} \mid \Sigma] = \begin{cases} 1 & i = j,\ s = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\mathrm{Var}[(A_s)_{ij} \mid \Sigma] \propto \frac{\lambda^2 \psi_j}{s^{2\alpha}} \cdot \frac{\sigma_{ii}}{\sigma_{jj}}$$

Interpretation

- Each series is assumed to be persistent in its own first lag.
- Cross-variable lag effects are shrunk toward zero unless the data strongly support them.
- λ controls overall tightness, α controls lag decay, and ψ_j scales variable-specific uncertainty.

Why this fits our data

Volatility indices are persistent and correlated, so the Minnesota prior stabilises the system without ruling out spillovers.

Compact Theory: Learning Shrinkage from the Data

Hierarchical prior selection

$$p(\boldsymbol{\theta}, \gamma \mid \mathbf{y}) \propto p(\mathbf{y} \mid \gamma)\, p(\gamma), \qquad p(\mathbf{y} \mid \gamma) = \int p(\mathbf{y} \mid \boldsymbol{\theta}, \gamma) p(\boldsymbol{\theta} \mid \gamma)\, d\boldsymbol{\theta}$$

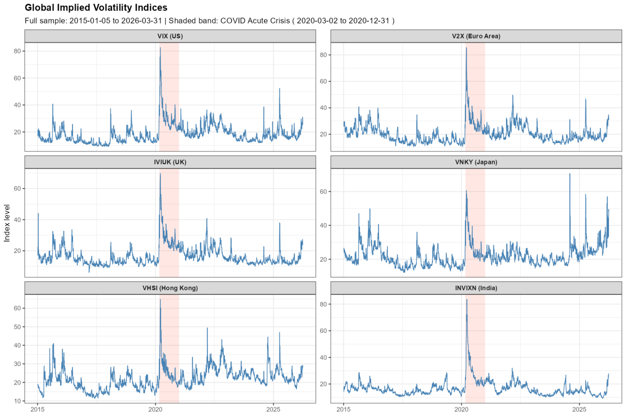
Interpretation

- Hyperparameters such as λ are not fixed manually.
- They are learned through the marginal likelihood.
- The procedure automatically balances fit against overfitting.

Why `BVAR` helps

- The package estimates the model, samples hyperparameters, and produces diagnostics in one workflow.
- Generalized IRFs, GFEVDs, and scenario forecasts all come from the fitted object and its posterior draws.
- This makes the methodology much easier to operationalise in practice.

Data and Empirical Setting



Setup

- VIX, V2X, IVIUK, VNKY, VHSI, INVIXN
- Sample: 2015-01-05 to 2026-03-31
- Calm sample: 2015-01-05 to 2019-12-31
- Crisis sample: 2020-03-02 to 2020-12-31

What the figure shows

- All markets spike sharply in early 2020.
- Levels differ by region, but common movement is strong.
- This makes the system well suited to a spillover analysis.

BVAR Package Workflow

```r
mn <- bv_minnesota(
  lambda = bv_lambda(mode = 0.2, sd = 0.4),
  alpha  = bv_alpha(mode = 2),
  psi    = bv_psi(scale = 0.004, shape = 0.004)
)

fit <- bvar(
  data = data_full, lags = 5,
  n_draw = 25000, n_burn = 10000,
  priors = bv_priors(hyper = c("lambda", "alpha", "psi"), mn = mn),
  irf = bv_irf(horizon = 20, fevd = TRUE, identification = FALSE),
  fcast = bv_fcast(horizon = 20)
)

irf_full  <- irf(fit, horizon = 20, conf_bands = c(0.05, 0.16))
gfevd_full <- compute_gfevd(irf_full, fit$sigma, var_names, horizon = 20)
pred_full <- predict(fit, conf_bands = c(0.05, 0.16))
```

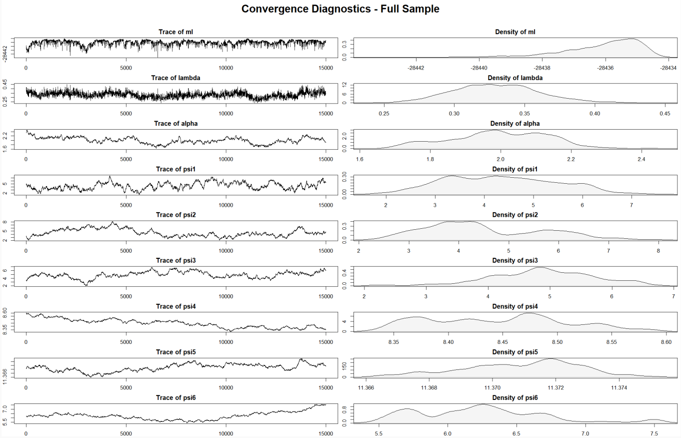
What this workflow gives us

- One object for estimation, diagnostics, and post-estimation
- Generalized IRFs and GFEVD-ready posterior draws
- Conditional forecasts for stress scenarios

Why this matters

The package makes the full Bayesian workflow transparent enough for teaching and gives us the objects needed for ordering-invariant post-estimation analysis.

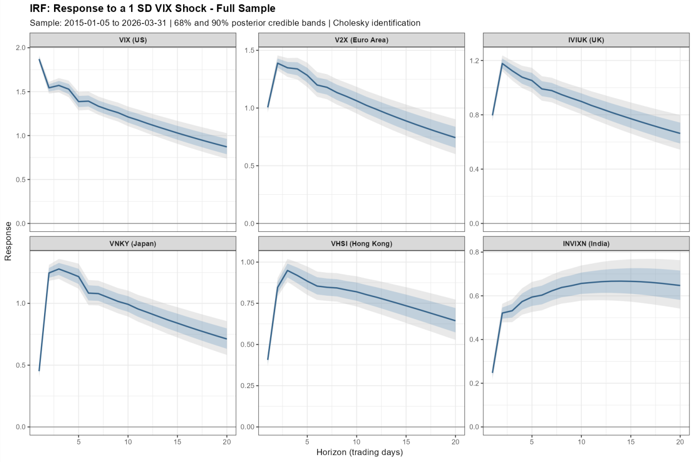
Minnesota-Only Convergence Diagnostics



Takeaway

- λ and α stabilise in clear regions.
- Some variable-specific scales mix more slowly, which is normal in a six-variable hierarchical system.
- Overall, the baseline Minnesota specification looks usable.

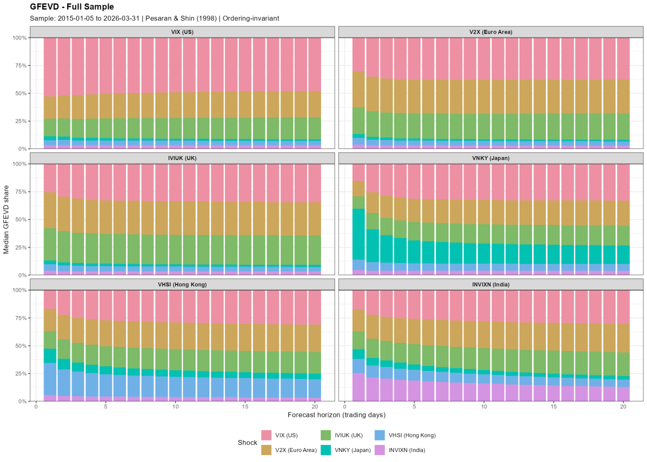
Full-Sample Generalized IRFs: 1 SD VIX Shock



What we learn

- Without imposing a recursive ordering, every market still reacts positively to a VIX shock.
- Europe, the UK, and Japan respond strongly on impact.
- India reacts more gradually, but the response remains positive over the full horizon.
- The slow decay is consistent with persistent volatility dynamics.

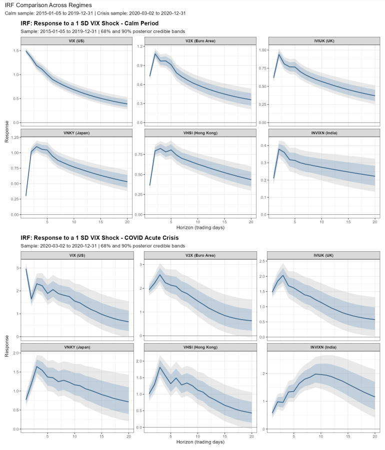
Full-Sample GFEVD



What we learn

- The ordering-invariant GFEVD still points to VIX as the main external shock in most panels.
- V2X and IVIUK remain especially exposed to US volatility.
- Japan, Hong Kong, and India keep more mixed local and regional contributions.
- GFEVD complements the generalized IRFs by showing which shocks matter quantitatively without imposing causal ordering.

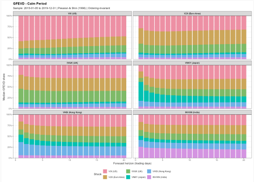
Crisis vs Calm: Generalized IRFs

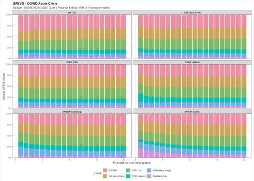


What changes in crisis?

- Crisis responses are larger on impact and more uncertain.
- Europe and the UK react much more strongly during 2020-03-02 to 2020-12-31 than in 2015-01-05 to 2019-12-31.
- India becomes far more responsive and peaks later, suggesting delayed but substantial transmission.
- The positive sign survives without imposing a recursive ordering, but the magnitude depends strongly on market conditions.

Crisis vs Calm: GFEVD





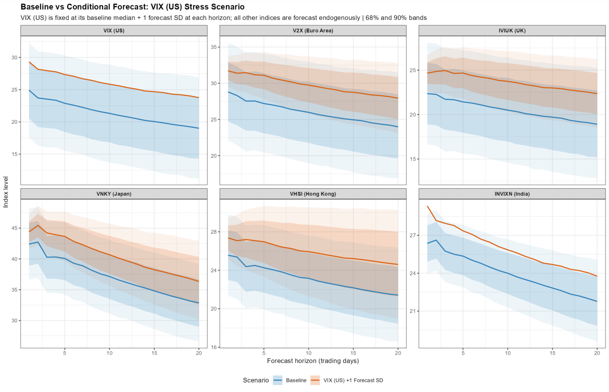
What changes in crisis?

- The crisis period shifts Hong Kong and India away from own or regional shares toward VIX, V2X, and IVIUK.
- Japan also becomes less self-driven as global shocks take more of the variance.
- VIX remains important, but the crisis decomposition becomes more broadly global rather than locally concentrated.

Interpretation

Ordering-invariant variance shares show broader contagion in crisis periods: local components shrink and common external stress explains more of the system.

Conditional Forecast: Persistent VIX Stress



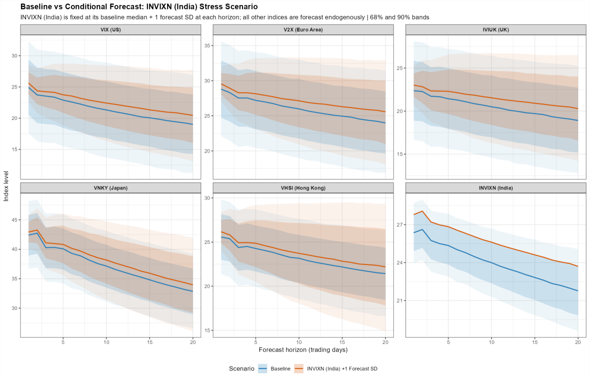
What we learn

- Fixing VIX one forecast standard deviation above baseline lifts every other forecast path.
- V2X, IVIUK, and VNKY show the largest upward shifts.
- The gap to baseline persists over the full horizon, so this is not just a one-day response.

Why this showcases the package

The scenario is implemented directly with `bv_fcast()` and `predict()`, so stress testing becomes part of the standard workflow.

Conditional Forecast: India Stress Scenario



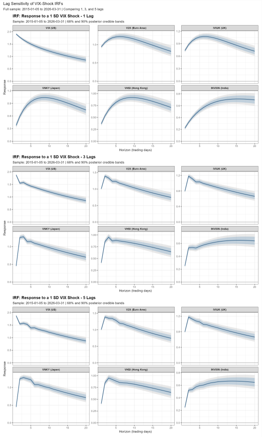
What we learn

- A one-standard-deviation India shock still raises the other paths, but much less than the VIX scenario.
- This is interesting because India is the only emerging-market index in the system.
- The figure suggests weaker outward transmission from India than from the US.

Interpretation

India is affected by global volatility, but its own stress scenario appears to be less system-defining than a comparable VIX shock.

Robustness to Lag Length



What we learn

- The ranking of responses is stable across 1, 3, and 5 lags.
- Shorter lag models smooth the early dynamics.
- Five lags allow sharper short-run spillovers, but the economic message stays the same.

Bottom line

Our conclusions are not an artefact of one particular lag choice.

SOC and SUR: What Do They Add?

Sum-of-coefficients prior (SOC)

$$y^{+} = diag\left(\frac{\bar{y}}{\mu}\right), \qquad x^{+} = [\mathbf{0}, y^{+}, \ldots, y^{+}]$$

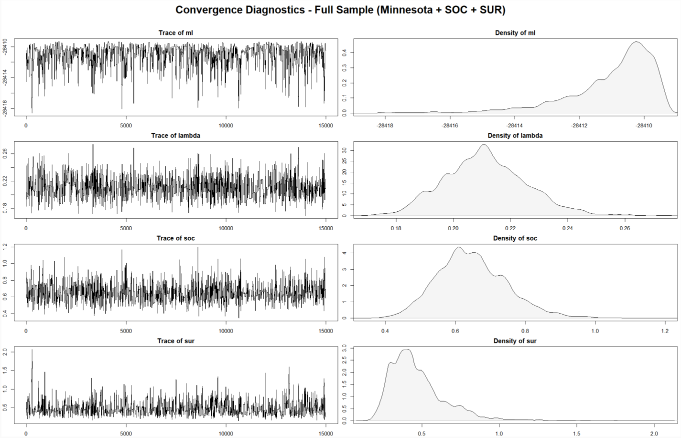
Single-unit-root prior (SUR)

$$y^{++} = \frac{\bar{y}^{\top}}{\delta}, \qquad x^{++} = \left[\frac{1}{\delta}, y^{++}, \ldots, y^{++}\right]$$

Interpretation

- SOC pushes the system toward persistent no-change dynamics.
- SUR allows common stochastic trends or cointegration-like persistence.
- Because our series are volatility levels, these priors are admissible as a robustness extension.

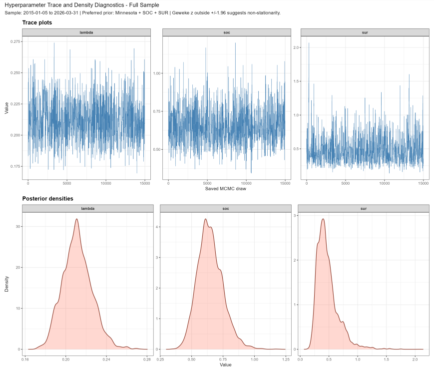
Minnesota + SOC + SUR: Convergence Diagnostics



Takeaway

- The marginal likelihood, λ, SOC, and SUR fluctuate around stable regions.
- Posterior densities are reasonably concentrated.
- The extended specification also converges in a usable way.

SOC/SUR Traces and Why the Results Barely Change



What the traces show

- The added hyperparameters stay in stable regions.
- Posterior mass concentrates around moderate values for λ, SOC, and SUR.

Why the economic results barely change

- The data are already persistent and estimated in levels.
- The Minnesota prior already captures most of the relevant persistence.
- SOC and SUR mainly reinforce that persistence rather than introduce a new cross-market story.

Project Conclusions

- A US volatility shock propagates positively to all markets in the system, so spillovers are clearly global.
- The transmission is stronger and more uncertain during the 2020 crisis window than in the calm period.
- The GFEVD still highlights VIX as the key external driver, while Hong Kong and India retain relatively larger local or regional components.
- Conditional forecasting confirms this asymmetry: a VIX stress scenario moves the whole system more than an India stress scenario.
- Our main findings are robust to lag choice and remain qualitatively unchanged when SOC and SUR are added.

What This Shows About the BVAR Package

- BVAR turns a full Bayesian workflow into a compact set of functions: prior setup, estimation, diagnostics, generalized IRFs, GFEVD-ready posterior objects, and scenario forecasts.
- The package makes hierarchical shrinkage operational rather than purely theoretical, which is especially valuable in medium-size VAR systems.
- It is strong not only for estimation, but also for communication: the same fitted object supports generalized responses, ordering-invariant variance shares, robustness checks, and stress scenarios.
- Our project is a good example of this strength: the package lets us move from model setup to empirical conclusions in a coherent and reproducible way.

Appendix: VAR Setup

Model

$$\boldsymbol{y}_t = \boldsymbol{a}_0 + A_1 \boldsymbol{y}_{t-1} + \cdots + A_p \boldsymbol{y}_{t-p} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(\mathbf{0}, \Sigma)$$

Why OLS is not enough

- Parameter count rises quickly with the number of variables and lags.
- Financial volatility series are persistent and strongly correlated.
- Unregularised VARs can overfit in-sample noise.

Why Bayesian shrinkage helps

- Priors shrink weak coefficients toward economically sensible benchmarks.
- The data can still dominate when the signal is strong.
- This improves stability and forecast performance in medium-size systems.

Appendix: Minnesota Prior

Prior mean

$$\mathbb{E}[(A_s)_{ij} \mid \Sigma] = \begin{cases} 1 & \text{if } i = j,\ s = 1 \\ 0 & \text{otherwise} \end{cases}$$

Prior variance

$$\mathrm{Var}[(A_s)_{ij} \mid \Sigma] \propto \frac{\lambda^2 \psi_j}{s^{2\alpha}} \cdot \frac{\sigma_{ii}}{\sigma_{jj}}$$

- λ: overall tightness
- α: lag decay
- ψ_j: variable-specific scaling
- Intuition: own first lags are persistent; cross-lag effects are shrunk strongly toward zero

Appendix: Hierarchical Hyperparameter Selection

Key idea

Treat the prior hyperparameters as random variables:

$$p(\boldsymbol{\theta}, \gamma \mid \mathbf{y}) \propto p(\mathbf{y} \mid \gamma)\, p(\gamma), \qquad p(\mathbf{y} \mid \gamma) = \int p(\mathbf{y} \mid \boldsymbol{\theta}, \gamma) p(\boldsymbol{\theta} \mid \gamma)\, d\boldsymbol{\theta}$$

Why this matters

- Shrinkage is learned from the data instead of fixed by hand.
- The marginal likelihood penalises overly flexible models automatically.
- Results become easier to justify and replicate.

How `BVAR` implements it

- Conjugate NIW structure gives closed-form posteriors for core parameters.
- Hyperparameters are sampled with Metropolis-Hastings.
- Diagnostics are available directly from the fitted object.

Appendix: SOC and SUR Dummy Priors

Sum-of-coefficients prior (SOC)

$$\boldsymbol{y}^{+} = diag\left(\frac{\bar{\boldsymbol{y}}}{\mu}\right), \qquad \boldsymbol{x}^{+} = [\boldsymbol{0}, \boldsymbol{y}^{+}, \ldots, \boldsymbol{y}^{+}]$$

Single-unit-root prior (SUR)

$$\boldsymbol{y}^{++} = \frac{\bar{\boldsymbol{y}}^{\top}}{\delta}, \qquad \boldsymbol{x}^{++} = \left[\frac{1}{\delta}, \boldsymbol{y}^{++}, \ldots, \boldsymbol{y}^{++}\right]$$

- SOC shrinks the system toward "no large changes from current levels."
- SUR allows persistent common movement and is useful when cointegration-like behaviour is plausible.
- These priors are designed for VARs in levels, which is why we only use them as a level-data robustness extension.

Appendix: Estimation Choices in Our Script

Baseline specification

- Six volatility indices in levels
- Lags: $p = 5$
- Draws: 25,000
- Burn-in: 10,000
- IRF / forecast horizon: 20 trading days

Identification and samples

- No recursive ordering is imposed.
- `bv_irf(..., identification = FALSE)`
- GFEVD follows Pesaran-Shin and is ordering-invariant.
- Calm sample: 2015-01-05 to 2019-12-31
- Crisis sample: 2020-03-02 to 2020-12-31

Interpretation note

All results should be read as reduced-form, ordering-invariant volatility transmission patterns rather than structural causal effects.

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